

BROAD AGENCY ANNOUNCEMENT
Operational Analysis, SSN/SSGN Survivability Program (S3P), and Hull,
Mechanical and Electrical (HM&E) Materials, Design Concepts, and
Technologies for Future Submarine Platforms

Broad Agency Announcement (BAA) Number: N00024-24-R-2120

Posting Date: 1 Nov 2023

Closing Date: 31 Oct 2026

Classification Code: A – Research and Development (R&D)

Contracting Office Address:

1333 Isaac Hull Avenue S.E., Washington Navy Yard, DC 20376

*Note: The terms “Navy,” “NAVSEA,” and “Government” in this BAA are one and the same.

Administrative Information

This publication constitutes a Broad Agency Announcement (BAA) as contemplated in Federal Acquisition Regulation (FAR) 6.102(d)(2) and 35.016. A formal Request for Proposal (RFP), solicitation, and/or additional information regarding this announcement will not be issued or further announced. This announcement will remain open for three (3) years from the date of publication unless otherwise extended. Initial responses to this announcement must be in the form of White Papers and may be submitted at any time during this three (3) year period. Proposals will be requested only from those Offerors selected as a result of the technical review of the White Papers made in accordance with the evaluation criteria specified herein. Contracts, or any other ordering vehicle identified as appropriate upon the receipt of the individual White Papers, may be awarded based on proposal evaluations and availability of funding.

The Naval Sea Systems Command (NAVSEA) will not issue paper copies of this announcement. NAVSEA reserves the right to select for proposal submission all, some or none from among the White Papers submitted in response to this announcement. Issuance of an RFP is not a commitment by NAVSEA to award a contract. NAVSEA provides no funding for direct reimbursement of White Paper or proposal development costs. Technical and cost proposals (or any other material) submitted in response to this BAA will not be returned. It is the policy of NAVSEA to treat all White Papers and proposals as competition sensitive information and to disclose their contents only for the purposes of evaluation.

Research Opportunity Title

SSN/SSGN Survivability Program (S3P), Hull, Mechanical, and Electrical (HM&E) Materials, Design Concepts, and Technologies for Future Submarine Platforms, and Operational Analysis.

Response Date

This announcement will remain open through the response date indicated on www.SAM.gov. White Papers may be submitted any time during this period.

1.0 Research Opportunity Description

The NAVSEA Advanced Submarine Systems Development (SUB 073) Office within the Program Executive Office for Attack Submarines (PEO SSN) hereby solicits UNCLASSIFIED White Papers through the use of a Broad Agency Announcement (BAA) (see FAR 35.016) for projects in the following areas:

1.1 SSN/SSGN Survivability Program (S3P) - A Navy objective for this BAA is to focus on technology ideas that can impact the survivability of submarines and unmanned undersea systems. The Navy is interested in White Papers for technologies with both near term (<10 years) and far term (>10 years) implications with concentration on 1) improving the understanding of the generation, radiation, propagation, and scatter of a variety of signal types (acoustic, chemical, optical, electromagnetic, radiological, hydrodynamic, and cyber) associated with submarine and unmanned system operations, to include both peacetime and wartime operations; 2) technologies that may mitigate these same phenomena; and 3) analytic tools, techniques, and processes which can be used to evaluate SS(G)N survivability against kinetic and non-kinetic affects during peacetime and wartime mission scenarios.

1.2 Hull, Mechanical, and Electrical (HM&E) Materials, Design Concepts, and Technologies for Future Submarine Platforms - A Navy objective for this BAA is to focus on technology ideas related to submarine HM&E materials, design concepts, and technologies for future submarine platforms. White Papers for technology ideas such as stealth technologies, coatings and hull treatments, naval architecture-related technologies that offer space and weight savings, cost reduction technologies, advanced energy storage, advanced propulsion, shafting and thrust technologies, composite components, payload handling and stowage technologies, payload launch and recovery technologies, advanced sensor technologies, advanced communication technologies, undersea and subsea & seabed warfare technologies, and survivability technologies are desired. Further technology development areas for which White Papers are requested include digital engineering techniques enabling platform design and integration of submarine advanced technologies.

1.3 Operational Analysis and Support for Undersea Warfare, Strategic Planning, and Wargaming Requirements – A Navy objective for this BAA is to focus on technical engineering analysis and research in support of the U.S. and allied nation submarines in the following areas:

- Studies, analyses and assessments addressing a broad variety of areas including but not limited to the Tactical Submarine Evolution Plan (TSEP) and Integrated Undersea Future Investment Strategy.
- Wargame design, development, planning, execution, and analysis in support of current and future undersea and seabed warfare capabilities and operations to support future technology decisions, capabilities and strategies.
- Modeling and Simulation (M&S) and other analytical capabilities to support mission engineering approaches or inform the development of strategic plans, capabilities, and mitigation options as well as identify technical and programmatic risks.
- Studies evaluating sensors, signatures, weapons systems, payloads, and platforms (both manned and unmanned), including the possible impact of new technology to in-service and future submarine operations in the presence of adversaries using kinetic and non-kinetic affects.

2.0 Proposal Selection Process and Contract Information

2.1 A two-phased (Phase I and Phase II) proposal selection process will be utilized under this BAA. This approach will save the Navy and Offerors considerable expense by not requiring development of full proposals unless necessary and desired by the Navy.

2.2 Phase I of the two-phased process consists of the submission of UNCLASSIFIED White Papers briefly describing the proposed technical services and/or technology, approach, experience, capabilities and past performance with a cost estimate for the applicable technology areas addressed in Section 1.0. White Papers are initially sought to preclude unwarranted effort on the part of the Offeror in also preparing Technical and Cost Proposals without an initial assessment of the technical and operational feasibility of the concept by the Navy. Offerors submitting White Papers may be invited to submit a Technical and Cost Proposal on all or part of their White Paper submission. However, any such invitation is in no way an obligation on the part of the Government to award a contract. Decisions regarding the invitation to submit a proposal will be based on the Navy's evaluation under Section 6.0. Furthermore, White Papers may be summarily rejected for the following reasons: 1) Proposal is not relevant to research interests identified in Section 1.0 above; 2) proposal was tried in the past or is in an existing technology; 3) proposal is high risk; and/or 4) proposal is unaffordable.

2.3 Phase II of the proposal selection process will consist of a request from the Government to the Offeror(s) for Technical and Cost Proposals to support the Navy in one or more of the technology areas listed in Section 1.0. These proposals can either be UNCLASSIFIED or CLASSIFIED. Technical proposals should include maturing the technology leading to laboratory and at-sea testing, and participating in technical/program reviews. An Offeror can be considered for contract award only upon preparation and submittal of a fully compliant Phase II Proposal. In the event that a Phase II Technical and Cost Proposal is requested, the applicable request for proposal will contain the terms and conditions and clauses to be incorporated in the potential contract.

2.4 Any contract(s) awarded based on responses to this BAA are considered to be the result of full and open competition (see FAR 6.102(d)(2)(i)) and in full compliance with the provisions of Public Law (PL) 98-369, "The Competition in Contracting Act of 1984."

2.5 The planned schedule for this BAA is as follows: This BAA will be open for a three (3) year period for Phase I, during which time White Papers may be submitted. Offerors are encouraged to submit White Papers as soon as practical within the three (3) year period for each applicable technology area. Offerors may submit multiple White Paper topics. If an Offeror submits multiple White Papers, each shall be submitted in response to a different topic or explore different technologies within one topic. The Navy intends to use its best efforts to complete its review of White Papers within approximately two (2) months of receipt of each individual White Paper. Any Offerors invited (as appropriate) to submit Technical and Cost Proposals shall do so as directed by the invitation (see Section 2.7 below). NAVSEA anticipates that such Offerors will usually be instructed to submit proposals within one (1) month of receiving the invitation to submit from the Navy. The evaluation of proposals, including discussions/negotiations as required, will be initiated approximately three (3) months after Proposals have been received. The Navy anticipates that the earliest potential contract award(s) made as a result of this BAA will be in the 1st Quarter of Fiscal Year (FY) 24. The Government reserves the right to award contracts after the closing date of this BAA.

2.6 Offerors are advised that only the Contracting Officer is legally authorized to bind the Government. The Prime Points of Contact (POCs) for submission of all Phase I White Papers and Phase II Technical and Cost Proposals, as well as Contractual questions are Ms. Elizabeth Feliciano, Contracting Officer, Mr. David Kinney, Contract Specialist and Mr. Kerry Payne, Contract Specialist. Their contact information may be found in Section 7.0.

Notes:

1. All UNCLASSIFIED communications pertaining to Section 1.0 shall be submitted via e-mail.
2. The intent to ask a CLASSIFIED question related to Section 1.0 shall be directed to Ms. Elizabeth Feliciano, Mr. David Kinney and Mr. Kerry Payne, who will arrange for the CLASSIFIED question to be asked through a secure method of communication.

2.7 Those Offerors selected to proceed to Phase II of the selection process will be notified by the Contracting Officer. The invitation to submit a Phase II proposal will include a date and time for submittal.

3.0 White Paper Preparation Costs

3.1 Offerors are hereby advised that the Government will not directly pay for any White Paper preparation costs resultant from this BAA. The cost of preparing White Papers is NOT an allowable direct charge to any resulting contract that may be awarded as a result of this announcement, or any other contract with the Government.

4.0 Data Rights & Proprietary Data

4.1 The Government desires a minimum of Government Purpose Rights in technical data and computer software delivered under contracts resulting from the two-phased selection process described herein.

4.2 The Government will treat White Papers the same as bid and proposal information (see DFARS 252.227-7016). White Papers and Technical and Cost Proposals containing data that the Offeror does not want to be disclosed to the public for any purpose, or used by the Government except for evaluation purposes, shall be marked as follows:

- For White Papers –

(1) Mark the cover page with the following legend:

“This document includes data that shall not be disclosed outside the Government and shall not be duplicated, used, or disclosed – in whole or in part – for any purposes other than evaluation. If, however, a contract is awarded to this Offeror as a result of – or in connection with – the submission of this data, the Government shall have the right to duplicate, use, or disclose this data to the extent provided in the resulting award document. This restriction does not limit the Government's right to use information contained in this data if it is obtained from another source without restriction. The data subject to this restriction are contained in sheets marked "Proprietary" or that contain the legend prescribed by FAR 52.215-1.” And

(2) Mark each sheet of data that that the Offeror does not want to be disclosed to the public for any purpose, or used by the Government except for evaluation purposes with the following legend:

“Use or disclosure of data contained on this sheet is subject to the restriction on the cover page of this proposal.”

- For Technical and Cost Proposals –

(1) Mark the title page with the following legend:

“This proposal includes data that shall not be disclosed outside the Government and shall not be duplicated, used, or disclosed-in whole or in part-for any purpose other than to evaluate this proposal. If, however, a contract is awarded to this offeror as a result of-or in connection with-the submission of this data, the Government shall have the right to duplicate, use, or disclose the data to the extent provided in the resulting contract. This restriction does not limit the Government's right to use information contained in this data if it is obtained from another source without restriction. The data subject to this restriction are contained in sheets [insert numbers or other identification of sheets].” And

(2) Mark each sheet of data it wishes to restrict with the following legend:

“Use or disclosure of data contained on this sheet is subject to the restriction on the title page of this proposal.”

5.0 Submission Information

5.1 The Government reserves the right to select all, some, or none of the White Papers in response to this announcement. Offerors must be able to certify that they have an appropriate facility clearance to meet the security requirements of work proposed, and key personnel must be certified as holding an appropriate clearance. The security clearance level for Technical and Cost Proposals is up to and including a TOP SECRET (TS) Sensitive Compartmented Information (SCI) level.

5.2 White Papers submitted under this BAA shall be UNCLASSIFIED. Technical and Cost Proposals submitted shall be UNCLASSIFIED to the greatest extent possible.

5.3 White Papers submitted in response to Section 1.0 of this BAA shall follow the instructions outlined in 5.4.

5.4 White Papers (Phase I): UNCLASSIFIED White Papers

5.4.1 White Paper Format

The page limit for the White Papers shall not exceed seven (7) pages (excluding cover page and resumes). Paper size shall be no larger than 8.5” x 11” paper with no less than 1” margins on all sides. The font shall not be less than 12 point. The files shall be in a format that is compatible with and capable of being opened in the most current Microsoft Office Suite or Adobe Acrobat. All spreadsheets shall be in Microsoft Excel format such that each mathematical equation is obvious and traceable. All files submitted to the Government must be scanned to ensure that there are no computer viruses within the file. No fields or documents shall be password protected by Offerors. If files are compressed, the necessary decompression program must be included.

White Papers shall be submitted via e-mail to:

- Elizabeth Feliciano, Contracting Officer at Elizabeth.feliciano-hernandez.civ@us.navy.mil,
- David Kinney, Contract Specialist at david.g.kinney4.civ@navy.us.mil, and
- Kerry Payne, Contract Specialist, kerry.l.payne3.civ@us.navy.mil.

5.4.2 White Paper Content

a. The cover page shall be labeled “WHITE PAPER,” and shall include the BAA number, proposal title, Offeror information, CAGE code, SAM Unique Entity Identifier (UEI) number, and technical and administrative points-of-contact along with e-mail addresses, and telephone numbers;

b. Executive Overview of the proposed technology improvement area as well as a summary of any innovative claims, and the Offeror's experience, capabilities and past performance in this area;

c. Technical Description, which shall be sufficiently accurate and complete, and consist of discussions of the technology improvement(s) being proposed and the ingenuity associated with the concept. If any portion of the proposed effort is predicated upon the use of Government-owned resources of any type, the Offeror shall clearly identify: the resources required, the date the resource is required, the duration of the requirement, the source from which the resource may be acquired, if known, and the impact on the effort if the resource is not provided. This section shall specifically address any intellectual property claims associated with the outcome of this effort. Specifically, the Offeror shall justify the use of proprietary, vendor-unique, or closed components including but not limited to hardware, software and interfaces in current or future designs;

d. Offeror's capabilities and past performance including a list of related and/ or relevant projects or contracts of a similar nature to the technologies proposed, to include contract numbers and Government point of contact, if applicable;

Notes:

- The Government reserves the right to use contract/project information provided in the Offeror's White Paper to obtain data from other sources, such as the Contractor Performance Assessment Reporting System (CPARS) database and from the points of contact identified by the Offeror in its White Paper.
- Where the Offeror does not have recent or relevant past performance information, the Offeror shall affirmatively state that it does not have recent and relevant past performance information for the Government to consider.

e. Resumes of key personnel;

f. A high-level schedule for the proposed effort;

g. A brief discussion of the overall approach to management of this effort. This shall include the organization, project/function/subcontractor relationships, Government interfaces, and how performance, cost and schedule will be assessed and controlled;

h. Draft Statements of Work (SOW) including deliverables (including data associated with the proposed technology) and key milestones, by fiscal year, to accomplish the effort proposed in the White Paper;

Note: The Navy reserves the right to negotiate the contents of the SOW if it desires to pursue contract award in Phase II;

i. Cost Estimate on a fiscal year basis (October – September); and

j. Identification of all technical data and computer software that the Offeror anticipates asserting should be furnished to the Government with less than unlimited rights should a Phase II proposal be requested. In addition to identifying all such technical data and computer software, identify the basis for the assertion and the anticipated asserted rights category.

5.5 Technical and Cost Proposals (Phase II): If a Technical and Cost proposal is solicited by the NAVSEA Contracting Officer, the proposal content requirements will be specified in the RFP.

6.0 Evaluation

6.1 The selection of White Papers submitted in response to this BAA will be based on a technical and peer review process. The major purpose of the evaluation will be to determine the relative merit of the technical approach of each White Paper. Business and contractual aspects, including proposed cost estimates and budgetary constraints, will also be considered as part of the evaluation. The selection of White Papers to continue through to Phase II of this process will be based on the potential benefits to the Government weighed against the cost estimates provided in the White Papers, in view of the availability of funds.

6.2 If White Papers are not summarily rejected for the reasons cited in subparagraph 2.2 above, then evaluation will be based on the following criteria (listed in descending order of importance for evaluation purposes).

1. Technical merit and approach;
2. Experience and capabilities of the Offeror, and experience of the key personnel who are critical to program success;
3. Past Performance;
4. Cost estimates for budgetary feasibility and;
5. Data Rights Assertions.

Note: Each White Paper will be evaluated against the above criteria and not against each other since proposals are not submitted in accordance with a common work statement.

IMPORTANT NOTE: The Government intends to use Government personnel for the evaluation of White Papers and proposals from activities other than the Naval Sea Systems Command or Program Executive Office for Attack Submarines. The Government also reserves the right to use contractor support personnel to assist in providing technical advisory support regarding any White Papers and full cost proposals and technical proposals (if requested) ensuing from this BAA. In addition, the Government reserves the right to use selected scientific and engineering expert personnel from Johns Hopkins University Applied Physics Laboratory (JHU/APL), Advanced Research Laboratory at the University of Texas (ARL/UT), Massachusetts Institute of Technology Lincoln Laboratory (MIT/LL), and Pennsylvania State University Applied Research Laboratory (PSU/ARL). These engineering experts will assist in providing their respective technical expertise, as

applicable, in the evaluation of any White Paper and full cost proposal and technical proposal (if requested) ensuing from this BAA. As required, the additional government evaluators, contractor support personnel, and engineering experts will be bound by Non-Disclosure Agreements to protect proprietary and source-selection information. Additionally, the contractor support organizations described above are prohibited from submitting White Papers in response to this BAA.

Should the Offeror have any objections to personnel from these organizations participating in the evaluation of White Papers and/or proposals, written notification shall be provided to the Contracting Officer within ten (10) business days prior to the Offeror's submission of White Paper(s). If there is an objection, the specific reason(s) for the objection(s) shall be stated in the written notification, and the specific non-government laboratory or support contractor of concern shall be identified.

6.3 One (1) or more awards may be made under this BAA to the same Offeror based on the evaluation criteria above and the availability of funds.

6.4 White Papers may be selected to proceed to tasking under existing contracts, if active and with related scope, thereby eliminating the need for Phase II proposals.

7.0 Points of Contact

- Contracting Officer: Ms. Elizabeth Feliciano, SEA 02212, Elizabeth.feliciano-hernandez.civ@us.navy.mil, (202) 372-5848
- Contract Specialists:
 - Mr. David Kinney, SEA 02212K, david.g.kinney4.civ@navy.us.mil, (202) 781-0135
 - Mr. Kerry Payne, SEA 02212P, kerry.l.payne3.civ@us.navy.mil, (202) 781-2894